Quantitative assessment of collagen hydrogel stiffness under varying polymerization conditions

Biology Practicals

RAUCH Clément, SOFRONIOU Antriana

Abstract

The aim of this project is to determine the stiffness of collagen hydrogels under varying polymerization conditions. Collagen hydrogels are particularly interesting to study because they serve as biologically relevant 3D scaffolds for cell culture, providing an environment that mimics in vivo tissue architecture in a better way than traditional 2D substrates. However, these gels are also challenging to handle and to mechanically characterize due to their soft nature and their sensitivity to preparation conditions. The goal of this experiment is to find a simple and reliable way to quantify their stiffness. To do so, we compare the Young's modulus of gels prepared with different collagen concentrations. We used two methods. In the first experimental set-up, we compute the Young's modulus (E) from the shear modulus (G) measured by a shear force machine. In the second experimental set-up, we measure gel deformation caused by glass beads to measure Young's modulus (E). Our findings yield that the shear force machine is unreliable because of polymerization issues and other experimental problems whereas the second experimental set-up produces reasonable and consistent results for the stiffness of collagen hydrogels. Further improvements should focus on the gel preparation as it was a major source of errors.

Contents

1 Introduction **2**
1.1 Collagen hydrogels . . . 2
1.2 Relevance of our experiment . . . 2

2 Methods **2**
2.1 Preparation protocol of hydrogels . . . 2
2.2 First experimental set-up . . . 3
2.2.1 Quantifying the stiffness . . . 3
2.2.2 The shear force machine . . . 4
2.2.3 Collecting data . . . 4
2.3 Second experimental set-up . . . 5
2.3.1 The Hertz contact model . . . 5
2.3.2 Collecting data . . . 6

3 Results **6**
3.1 First experimental set-up . . . 6
3.2 Second experimental set-up . . . 7

4 Discussion **7**
4.1 First experimental set-up . . . 7
4.2 Second experimental set-up . . . 8

5 Conclusion **9**

References **10**

Appendix **11**
Section A . . . 11
Section B . . . 12
Section C . . . 14

1 Introduction

This study aims to quantitatively assess the stiffness of collagen hydrogels under varying polymerization conditions. By comparing results using different methods, we also aim to determine the simplest way to quantify the stiffness of these gels. After introducing the topic, we detail the methodology used in our experiments. Then, we present the results we obtained. Lastly, we discuss these results and conclude.

1.1 Collagen hydrogels

Collagen is the most abundant protein in the human body, making up about one-third of its total protein content. In fact, collagen is one of the major structural component of the extracellular matrix (ECM), a three-dimensional network of macromolecules surrounding cells in connective tissues, where it provides strength, elasticity, and support. Indeed, collagen molecules form long fibrils that give tissues their mechanical stability and resilience. The body naturally produces collagen, but its synthesis decreases with age, leading to weaker tissues and visible signs of aging such as wrinkles and joint stiffness. In addition, there are many types of collagen, but type I is by far the most abundant and can be found in the skin, tendons, ligaments, and bones.

When a solution of collagen is neutralized (pH $\approx$ 7.4) and warmed to physiological temperature (T $\approx$ 37 °C), the molecules self-assemble into collagen fibrils through polymerization. This process leads to the formation of a three-dimensional structure known as a hydrogel. Collagen hydrogels are bio-compatible and support cellular processes such as stem cell differentiation. In terms of mechanical properties, they are soft materials, with stiffness values typically between 10 Pa and 1 kPa.

1.2 Relevance of our experiment

According to studies [1], the development, mobility, and expression of cells differ when they are cultured in standard 2D petri dishes rather than a 3D matrix. In particular, when derived from natural polymers such as proteins, hydrogels can mimic the ECM found in human tissues, and collagen is one of the most commonly used components for this purpose. Therefore, if the reconstruction and reproducibility of such collagen hydrogels is achieved, then they can be used in experiments replacing petri dishes giving more accurate results.

However, collagen-based hydrogels are difficult to fabricate reproducibly. Their microstructure and mechanical properties are highly sensitive to small variations in the preparation protocol, such as polymer concentration, pH, ionic strength, and temperature. For this reason, each fabrication protocol must be carefully characterized to ensure that the desired material properties are consistently achieved. Moreover, the stiffness of the collagen hydrogel plays an essential role, as it affects how cells attach, spread, and interact with their surroundings. Thus, if the hydrogel is too soft or too stiff, cells may not respond in a physiologically relevant way. Consequently, determining the right protocol to measure the stiffness of the gel is very particularly relevant for further biology experiments.

2 Methods

The stiffness of a collagen hydrogel is characterized by the Young's modulus. Therefore, our methods focus on its determination. In fact, during the lab session, we worked with two different experimental setups. In the first setup, the Young's modulus was determined by the measurement of the shear force. In the second setup, the Young's modulus was determined by the measurement of the deformation of the gel after a bead has fallen on it. To prepare the gel, we followed a specific protocol common to the two setups.

2.1 Preparation protocol of hydrogels

In this section we detail the material used, the reagents and the protocol to prepare the collagen hydrogels.

1. **Materials**
   - Pipettes with volume ranging from 0.5 μL to 1000 μL
   - Pipette tips
   - 1.5 mL Eppendorf tubes
   - Gloves, lab coat
   - Waste bin

2. **Reagents**
   - Collagen
   - Neutralizing solution
   - Hydrochloric acid (HCl)
   - Distilled water
   - (Optional) Dye
   - Ethanol (for cleaning)

To obtain the collagen hydrogel, we mix a solution of collagen with a neutralizing solution in an acidic environment as shown in Figure 1. We also add some drops of dye to be able to better see and monitor the gel during the experiment.

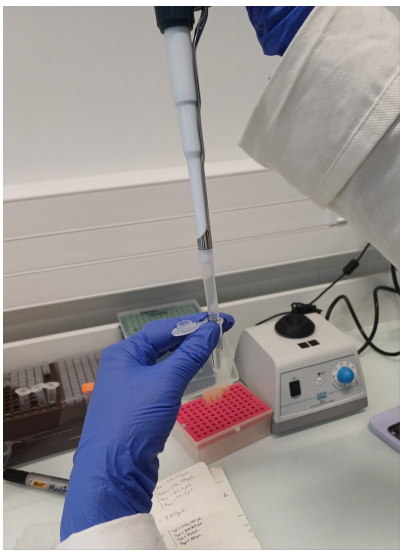


Figure 1: Insertion of collagen to the solution

Determination of the reagents' volumes

Depending on the desired collagen concentration and final volume, different volumes of reagents' solutions are used. To this end, in Appendix A, the general protocol can be found. Here, we will explicitly describe the protocol for a $100\mu L$ hydrogel with 4% collagen concentration. Notice that the collagen concentration in the solution of collagen is $c_{i,\text{collagen}} = 6.2mg/mL$, yet we want to dilute it to $c_{f,\text{collagen}} = 4.0mg/m$. We also require the neutralization solution to be 6.8% of the final volume and the final volume to be $V_{total} = 100$. Hence, we find the following volumes for each reagent:

- $V_{collagen} = 64.5\mu L$
- $V_{HCl} = 25.5\mu L$
- $V_{NS} = 6.8\mu L$
- $V_{H_2O} = 3.2\mu L$

Protocol

To prepare the hydrogel we follow the protocol described below:

1. Place the Eppendorf tube on ice to keep it cold to slow down the reaction.
2. Start by adding the collagen carefully to make sure all of it went in the tube and it was not stuck in the tip since it is very viscous.
3. Add the HCl and mix well.
4. Add the distilled water and the dye and mix well.
5. Finally add the neutralization solution and mix again.
6. In case that there is a significant amount of bubbles, it is possible to centrifuge the sample for a very short time to eliminate the bubbles, but keep in mind that it may harm the collagen if it lasts too long.
7. Cast the prepared mixture on mold of the set-up.
8. Place the set-up with collagen hydrogel in the incubator preheated at 37°C (or 30°C).
9. Leave it in the incubator as long as it needs to set. For an incubator at 37°C, 15 minutes are sufficient. For an incubator at 30°C, more than 30 minutes are required.
10. If the gel has not set, wait another 10 minutes, then check if it has set. If it is not the case, repeat until it has set.
11. Lastly, remove the gel from the incubator. The gel is ready for measurements.

2.2 First experimental set-up

The stiffness is measured by shearing the sample using a custom-made setup and collecting data through a dedicated software named Agnes.

2.2.1 Quantifying the stiffness

The shear modulus G measures stiffness in shear. Thus, it expresses the extent to which a material deforms under shear stress. It is defined as the ratio of the shear stress σ to the shear strain ϵ:

$$G = \frac{\sigma}{\epsilon}.$$

Recall that $\sigma = \frac{F}{A}$, where F is the applied shear force and A is the cross-sectional area over which

the force is applied. Also recall that $\epsilon = \frac{\Delta x}{l}$, where Δx is the lateral displacement produced by the force, and l is the distance over which the force acts. Therefore, one obtains that:

$$G = \frac{Fl}{A\Delta x}.$$

The above equation can also be rewritten using the volume V and the height h as follows:

$$G = \frac{Flh}{V\Delta x}.$$

To obtain Young's modulus E one measures the stiffness in tension or compression. It is related to the shear modulus G via the formula:

$$E = 2G(1 + \nu).$$

For collagen hydrogels, we take $\nu = 0.3$ [2].
It follows that if we are able to measure the shear modulus for each collagen hydrogel, we can determine its Young's modulus, and hence quantify its stiffness.

2.2.2 The shear force machine

This shear force machine comprises two plates (made with two glass cover-slips and a base to hold them), a screw that acts on the upper surface of the plate, and sensors that connect to a computer. Pictures of the shear force machine is given in Figure 2.

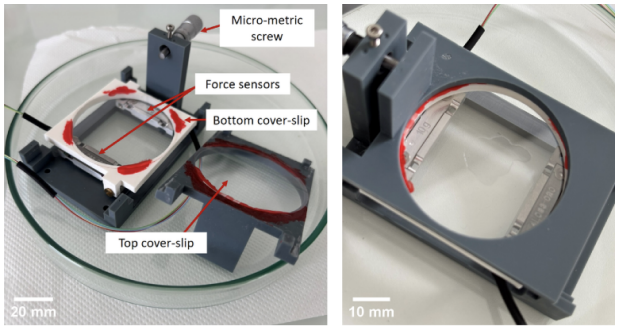


Figure 2: Shear force measurement setup.

To measure the distance between the two plates we used Optical Coherence Tomography (OCT), which is an imaging technique based on low-coherence interferometry that produces cross-sectional images. It works by detecting the echo time delay and intensity of light that is back-scattered from different depths. Using the OCT machine we measured the height between the plates to be $h = 443\mu m$.

2.2.3 Collecting data

One measurement consists of recording the shear force acting on the collagen hydrogel by turning the screw. An illustration of what the collagen hydrogel is expected to look like between the two plates is given by Figure 3.

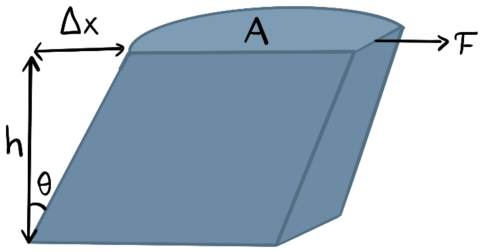


Figure 3: Illustration of hydrogel with the force F applied to it.

To this end, we used the Agnes software, which gave us the force. We then correlated the force to the amount of turns we screw, and obtained a graph of the force as a function of the displacement. A typical graph we managed to have in one of our first tests is given by Figure 4.

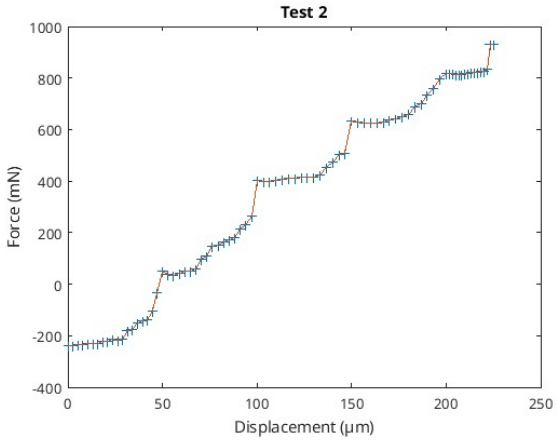


Figure 4: Force versus displacement for test 2 (with all data points)

We observe on the graph that the slope is not constant: the rate of increase of the shear force is increasing, reaches a peak per displacement and then relaxes before the next displacement starts. Therefore, the curve obtained is not linear: for this reason, we didn't retain this graph, but instead draw another graph where we plot the measurements of the force at the end of each displacement. We obtain Figure 5.

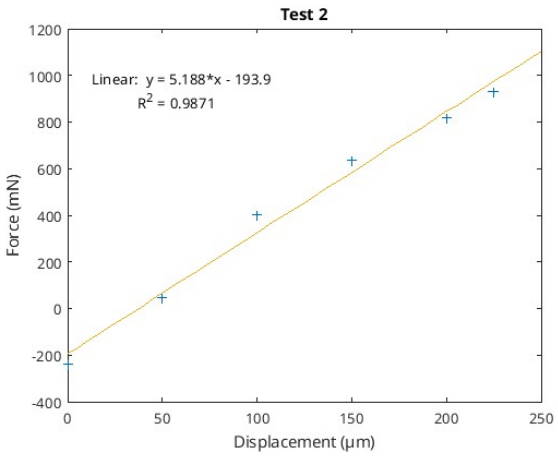


Figure 5: Force versus displacement for test 2 (with selected data points)

This graph is fully linear. Using the formula $G = \frac{Fl}{A\Delta x}$, and the slope of the plot $\frac{G}{\Delta x}$, we can determine the shear modulus G, and thus, the Young's modulus E. This quantifies the stiffness of the gel.

2.3 Second experimental set-up

In the second experimental set-up, the stiffness is determined by using the OCT to measure the deformation of the gel due to a micro bead falling from above. An example of a bead is given by Figure 6.

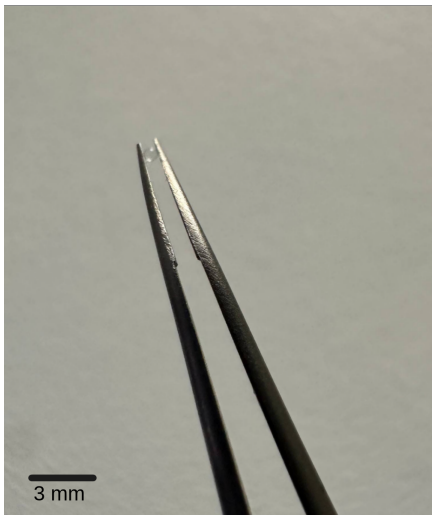


Figure 6: Bead to be placed on the hydrogel

The plates containing the gel are placed inside water under the OCT, meaning that we can consider directly the force created by a micro bead inside water without having to consider separately the water originally at the surface of the gel. A picture of the OCT is given by Figure 7. The lenses and the distance from the sample are adjusted manually.

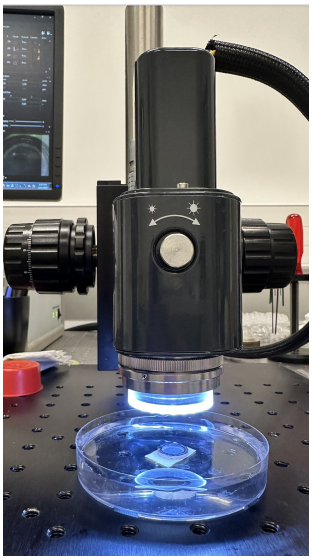


Figure 7: OCT machine emitting light

2.3.1 The Hertz contact model

To determine the Young's modulus, we used the Hertz contact model, which yields that the force F applied by the bead to the gel can be written as follows:

$$F = \frac{4R^{1/2}\delta^{3/2}E}{3(1-\nu^2)},$$

where F is the total force felt by the gel, ν the Poisson's ratio of the gel, R the radius of the beads, δ the displacement of the gel at the middle of the bead and E the Young's modulus of the gel. A schematic of the system is given by Figure 8.

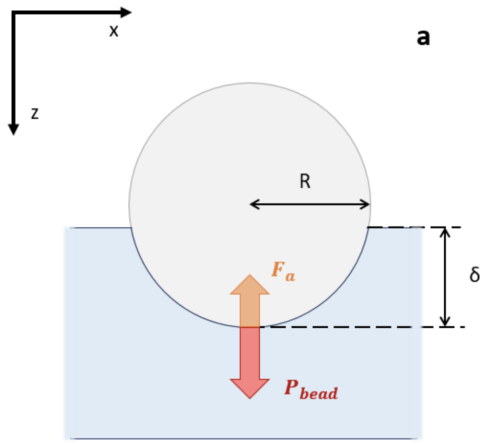


Figure 8: Schematic of the system.

Therefore, the Young's modulus is given by:

$$E = \frac{3F(1-\nu^2)}{4R^{1/2}\delta^{3/2}}$$

Yet, the total force consists in the sum of the weight of the micro-bead and its buoyant force:

$$\begin{aligned} F_a &= m_{bead} \cdot g - \rho_{water} \cdot g \cdot V \\ &= \frac{4}{3}\pi R^3 \cdot \rho_{glass} \cdot g - \rho_{water} \cdot g \cdot \frac{3R-\delta}{3}\pi\delta^2 \end{aligned}$$

The Poisson's ratio is considered to be $\nu = 0.3$.

2.3.2 Collecting data

To collect data from the measurements of the OCT, we used a dedicated software allowing to record sequences of images of the cross-sectional area. For each sequence of images obtained, using another software named Fiji, we selected the image where deformation was the biggest, and measured the bead radius and the deformation height on it. A typical image is given by Figure 9.

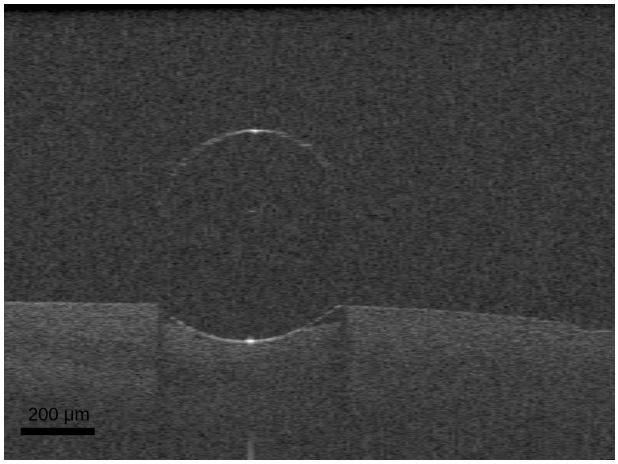


Figure 9: Cross-sectional area of the micro bead inserted in the collagen hydrogel

This process enables to determine the radius R of the bead and the deformation height δ for each micro bead inserted inside each collagen hydrogel prepared. Notice that, as the distances measured from the OCT are modified when the light passes through different media (here: water and glass), one gets that the actual distance d_a can be expressed as a function of the measured distance d_m as:

$$d_a = \frac{n_w}{n_g} d_w,$$

where n_a is the index of refraction of water and n_g is the index of refraction of glass. Therefore, the distance measured have to be corrected. This enables us to determine the total force, and hence, the Young's modulus, giving us the possibility to quantify the stiffness of each collagen hydrogel depending on the collagen concentration.

3 Results

In this section, we detail our results for both experimental setups.

3.1 First experimental set-up

Firstly, notice that we prepared in total 20 hydrogels, but only 13 worked. Therefore, the success of polymerization was 65%. Due to problems with the shear force machine, we were only able to perform 11 measurements, but for 8 of them the borders of the mold were attained in the incubator and for 1 of them there was friction between the plates, meaning that the measurements were not valid. Therefore, in total, only 2 measurements could be successfully done out of the 20 hydrogels prepared, which means that the total success was 10%. Complete tables with all our measurements in each session can be found in Appendix B.

Overall, all our measurements were in the 10-100 kPa range. The slope of the graph obtained, the shear modulus calculated and the Young's modulus determined for the two successful measurements are given in Table 1. Therefore, the mean Young's modulus obtained with the shear force machine for a concentration of 4 mg/mL is 157.7 kPa.

Test number	11	12
Concentration (mg/mL)	4	4
Slope (N/m)	6170	5410
Shear modulus (kPa)	60.54	53.09
Young's modulus (kPa)	157.41	138.02

Table 1: Successful measurements for the first-experimental setup

In addition, even though these measurements are not reliable because the borders of the mold were attained in the incubator, we also notice an inverse correlation between the Young's modulus

and the volume for collagen hydrogels with concentration 1 mg/mL. This inverse correlation is given on the plot on Figure 10.

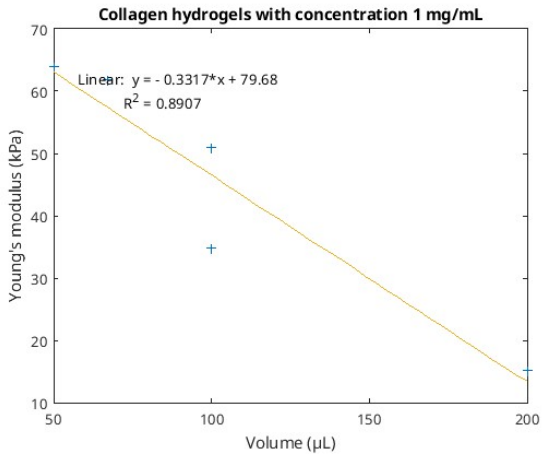


Figure 10: Young's modulus as a function of volume for collagen hydrogels with concentration 1 mg/mL

3.2 Second experimental set-up

Using the Hertz contact model, we measured the Young's modulus for hydrogels of different collagen concentrations. We faced the same difficulties obtaining polymerized gels as in the first experimental set-up, but a total of 7 gels were prepared. On average, we used 4 micro beads on each gel. For each bead, we measured the radius R of the bead and the deformation height δ of the hydrogel. We then computed the force and then the Young's modulus for each gel. The average Young's moduli obtained for each gel concentration tested is given in Table 2. The total average is found to be $E = 103.98$Pa. Detailed results for each gel and each test can be found in Appendix C.

In the following table are the average Young's modulus we computed for each gel can be seen.

C_{col} (mg/mL)	Young's modulus (Pa)
2	99.72
3	80.33
4	103.62
5	127.60
5.5	108.65

Table 2: Average Young's modulus for each collagen concentration for measurements using the OCT.

Plotting the results obtained in Table 2 yields Figure 11. We notice an overall increase of the Young's modulus as the concentration of the gels increases.

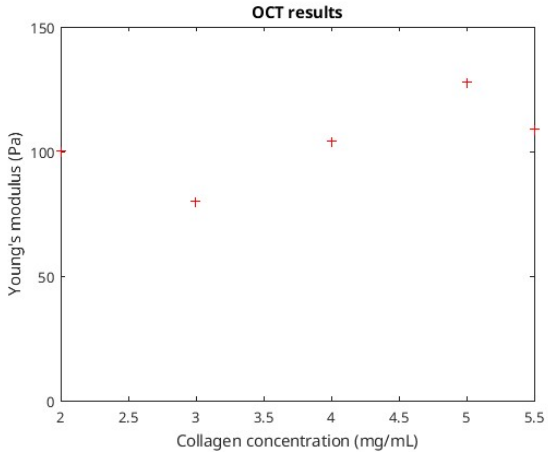


Figure 11: Young's modulus as a function of gel concentration for measurements using the OCT

4 Discussion

One of the most difficult task was the successful preparation of the hydrogel. Indeed, to obtain reproducible results, the concentration of collagen needed to be high enough and the volume of the gel needed to be well-chosen. In fact, at low concentration, the hydrogel remains very liquid and does not polymerize fully. In addition, we also had to minimize potential mistakes in the preparation of the hydrogel, which involved principally mistakes due to bad pipette calibration. Even though mistakes were minimized, this did not prevent gels to fail the polymerization process. A possible explanation for this could be the quality of the collagen and neutralization solution, which may have been damaged by time.

4.1 First experimental set-up

Concerning the volume of the gel, we noticed that for bigger volumes the boundaries of the plates were attained. Moreover, as mentioned earlier, there is a strong inverse correlation between the Young's modulus and the volume for collagen hydrogels with concentration 1 mg/mL. This inverse correlation heavily suggests the presence of friction forces, which adds up to the shear force. Indeed, in our method, friction forces contribute more to the total force for lower volumes than

for higher volumes, explaining why the Young's modulus is not constant with respect to the volume of gel.

Further errors in our measurements involve the assumption that the hydrogel is rectangular between the two plates. In reality since there is adhesion, meaning that water is attracted to the glass, the gel has a shape as illustrated in Figure 12. Therefore, the actual area that is sheared is greater than the one we measure from the geometry of the mold. To minimize this error, we can increase volume of the hydrogel as this reduces the proportion of the additional area sheared to the total area sheared. In this context, the volume of the gel had to be well-chosen. We retain that values for the volume between 20 μL and 100 μL are ideal.

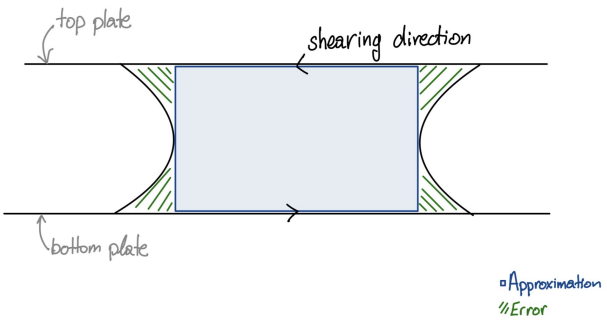


Figure 12: Illustration of the error we get from adhesion.

Another source of error is due to water coming out of the gel. This water, located in between the plates, was a factor that influenced the shear force. Finally, a last source of error is due to the cleaning process, which is not perfect. For this reason, leftovers of the previous gel remained on the plates and made the hydrogel slide more easily during the experiment, meaning it did not properly bind on both plates.

4.2 Second experimental set-up

The second experimental set-up allowed us to get more trustworthy measurements. It can be seen that our results were within $90 - 130$ Pa, which is in accordance with the 10 Pa $-$ 1 kPa range expected for collagen hydrogels. In particular, in a similar experiment conducted in the lab, the Young's modulus found was a bit less than the values we obtained. However, in this report, the experiment was not performed in water, resulting in the beads to slightly float as the collagen hydrogels contain water. This explains the difference in values and verifies that our results were logical.

In addition, the graph on Figure 11 demonstrates that the Young's modulus increases along with the collagen concentration of the hydrogels. This corresponds to what we would expect logically as the density of the gel increases with the collagen concentration. In fact, notice that the gel obtained for the concentration of $2mg/mL$ was not fully nor evenly polymerized. Indeed, approximately half of it was polymerized while the other half still remained quite liquid. Moreover, as depicted on Figure 13, the gel obtained for the concentration of $5.5mg/mL$ did not have a smooth surface, meaning that the areas where we could place the bead were limited, thus impacting the accuracy of measurements. Therefore, this can explain why the values for concentrations $2mg/mL$ and $5.5mg/mL$ don't follow the general trend of Figure 11.

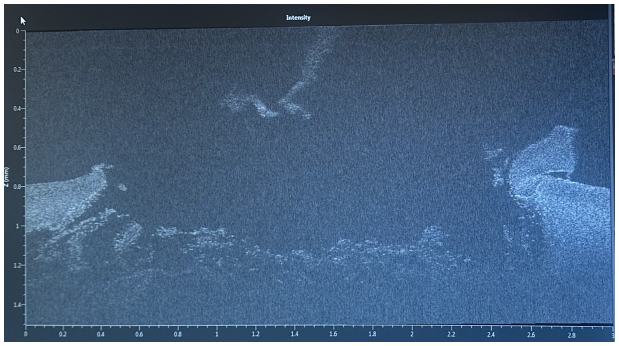


Figure 13: Image of the surface of the $5.5mg/mL$ concentration hydrogel in the $e_x - e_z$ plane

In addition, the following inaccuracies composed errors:

- **Refraction index.** The refraction index is set on the machine to be the one of water, but the light passes through air, water, glass and water again. Even though our method take that into account that it passes though water and glass in the computation of the distances, this can contribute to error to some extent.
- **Polymerization of the gel.** Some of the measurements were taken on gels that were not completely polymerized or for which the surface was not smooth enough.

- **Analysis of the data.** During the analysis of the data, human error was also present as we were manually choosing the correct cross-sectional area, as well as selecting the bead radius and deformation height.

5 Conclusion

In this experiment, we attempted to quantify the stiffness of collagen hydrogels of different concentrations using two methods. It appears from our results that, if it is true that collagen hydrogels have a huge potential when they are used as 3D cell-culture scaffolds, it is also true that obtaining reproducible polymerized gels is challenging as they are highly sensitive to parameters such as concentration, temperature and handling.

Using the first experimental set-up, we obtained values of Young's modulus in the 10-100 kPa range. However, these measurements suffered from systematic errors, including friction and gel leakage. As a result, two measurements were valid, making this method very difficult to put in place effectively in the long run. The results for that section are in no way reliable due to the lack of reproducibility. Using the second experimental set-up, we obtained more consistent and realistic results, with Young's moduli around 80-130 Pa. This suggests that the second method is a much more reliable approach for characterizing the properties of collagen hydrogels.

Our work demonstrates therefore that quantifying collagen hydrogel stiffness is perfectly possible, but a major limitation involves the reproducibility of the gels themselves. Further improvements should tackle the preparation of the gels. This said, the method of the OCT is definitely a promising technique for assessing collagen hydrogel stiffness in a simple and robust way.

References

[1] Duval, K., Grover, H., Han, L. H., Mou, Y., Pegoraro, A. F., Fredberg, J., & Chen, Z. (2017). Modeling physiological events in 2D vs. 3D cell culture. Physiology (Bethesda), 32(4), 266–277. https://doi.org/10.1152/physiol.00036.2016

[2] Cappello, J., d'Herbemont, V., Lindner, A., & du Roure, O. (2020). Microfluidic In Situ Measurement of Poisson's Ratio of Hydrogels. Micromachines, 11(3), 318. https://doi.org/10.3390/mi11030318

[3] RefractiveIndex.INFO. (n.d.). GLASS - BK7 optical glass (crown). Retrieved 9/12/2025, from https://refractiveindex.info/?shelf=3d&book=glass&page=BK7

Appendix

A. Volume calculations in the general collagen hydrogel preparation protocol

We define:

- C_i : concentration of the initial collagen solution (in our case $6.2 \frac{mg}{mL}$)
- C_{inter} : concentration of the intermediate collagen + HCl solution
- C_f : concentration of the final collagen + HCl + neutralizing solution (NS).
- V_{col} : Volume of collagen to be taken from the stock solution
- $V_{inter,col}$:volume of collagen in the intermediate collagen + HCl
- V_{NS} : volume of neutralizing solution required
- V_{HCl} : volume of HCl required
- V_{tot} : desired final volume
- β : proportion of collagen solution at C_f relative to the total gel volume
- V_{H_2O} : volume of distilled water added to complete the total volume

A dilution of the stock of collagen with HCl is performed using:

$$C_i \cdot V_{col} = C_f \cdot V_{tot} \iff V_{col} = \frac{C_f \cdot V_{tot}}{C_i}$$

The supplier specifies that 9 parts of collagen should have one part of neutralizing solution, which after dilution gives:

$$V_{inter,col} = 0.9 \cdot V_{tot}$$

$$V_{NS} = 0.1 \cdot V_{tot}$$

and:

$$V_{tot} = V_{inter,col} + V_{NS} = V_{col} + V_{HCl} + V_{NS}$$

Since the goal is to calibrate the final gel pH by varying the proportions of neutralizing solution and HCl (which are independent), we compute:

$$V_{HC\ell} = V_{tot} - V_{col} - V_{NS} = V_{tot} \left(1 - \frac{C_f}{C_i} - \alpha\right)$$

with $\alpha = \frac{V_{NS}}{V_{tot}}$.

We finally complete the volume with distilled water to avoid altering the pH and obtain the target volume.

Since the supplier indicated that 90% of the total mixture should be collagen, the proportion of neutralizing solution α is specifically determined for the collagen concentration C_f, and the remaining fraction $\beta - \alpha$ is added as distilled water. Thus:

$$V_{col} = \frac{C_{inter}}{C_i} \cdot V_{inter}$$

with $V_{inter} = V_{tot} \cdot (1 - \alpha - \beta)$ and $C_{inter} = \frac{C_f}{\beta}$. Hence:

$$V_{HCl} = V_{inter} - V_{col}$$

$$V_{NS} = V_{tot} \cdot \alpha$$

$$V_{H_2O} = V_{tot} \cdot (\beta - \alpha)$$

B. Results by session of the experiments using the first experimental setup

In total 20 gels were prepared for this experiment, but they did not all polymerize, meaning that some of the unsuccessful outcomes are mentioned here.

Date	2025-10-1
Test number	1
Collagen concentration (mg/mL)	Other protocol
Sample volume (μL)	157
Time in incubator (min)	15
Problems regarding the surface	Borders attained
Problems regarding the apparatus	Fine
Slope (N/m)	3400
Shear modulus (kPa)	4.25
Young's modulus (kPa)	11.05

Table 3: Measurements done on 1/10/2025 for the first experimental setup with 37°C for the temperature of incubator and 443 μm for the height between plates of the apparatus used.

Date	2025-10-8	2025-10-8	2025-10-8
Test number	2	3	4
Collagen concentration (mg/mL)	2	1	2
Sample volume (μL)	200	200	200
Time in incubator (min)	15	15	20
Problems regarding the surface	Borders attained	Borders attained	Borders attained
Problems regarding the apparatus	Fine	Fine	Fine
Slope (N/m)	4870	5980	6880
Shear modulus (kPa)	4.78	5.87	6.75
Young's modulus (kPa)	12.42	15.26	17.55

Table 4: Measurements done on 8/10/2025 for the first experimental setup with 37°C for the temperature of incubator and 443 μm for the height between plates of the apparatus used.

Date	2025-10-15	2025-10-15	2025-10-15
Test number	5	6	7
Collagen concentration (mg/mL)	1	1	1
Sample volume (μL)	100	100	67
Time in incubator (min)	15	15	15
Problems regarding the surface	Borders attained	Borders attained	Borders attained
Problems regarding the apparatus	Fine	Fine	Fine
Slope (N/m)	6810	9990	8130
Shear modulus (kPa)	13.36	19.61	23.81
Young's modulus (kPa)	34.75	50.97	61.92

Table 5: Measurements done on 15/10/2025 for the first experimental setup with 37°C for the temperature of incubator and 443 μm for the height between plates of the apparatus used.

Date	2025-10-22
Test number	8
Collagen concentration (mg/mL)	1
Sample volume (μL)	50
Time in incubator (min)	20
Problems regarding the surface	Borders attained
Problems regarding the apparatus	Fine
Slope (N/m)	6270
Shear modulus (kPa)	24.61
Young's modulus (kPa)	63.99

Table 6: Measurements done on 22/10/2025 for the first experimental setup with 37°C for the temperature of incubator and 443 μm for the height between plates of the apparatus used.

Date	2025-11-5	2025-11-5
Test number	11	12
Collagen concentration (mg/mL)	4	4
Sample volume (μL)	20	20
Time in incubator (min)	30	30
Problems regarding the surface	Fine	Fine
Problems regarding the apparatus	Fine	Fine
Slope (N/m)	6170	5410
Shear modulus (kPa)	60.54	53.09
Young's modulus (kPa)	157.41	138.02

Table 7: Measurements done on 5/11/2025 for the first experimental setup with 30°C for the temperature of incubator and 443 μm for the height between plates of the apparatus used.

Date	2025-11-12
Test number	15
Collagen concentration (mg/mL)	4
Sample volume (μL)	50
Time in incubator (min)	35
Problems regarding the surface	Fine
Problems regarding the apparatus	Friction between plates
Slope (N/m)	5160
Shear modulus (kPa)	103.20
Young's modulus (kPa)	268.32

Table 8: Measurements done on 12/11/2025 for the first experimental setup with 30°C for the temperature of incubator and 1000 μm for the height between plates of the apparatus used.

C. Summary of the experiments performed for the second experimental set-up

Sample	C_{col} (mg/mL)	Test	R_{bead} (μm)	δ (μm)	Force (N)	Young's modulus (Pa)
2	4	3	285	77.2	2.32×10^{-6}	1.39×10^{2}
3	4	1	277	109	2.09×10^{-6}	7.52×10^{1}
3	4	2	262	81.6	1.79×10^{-6}	1.03×10^{2}
3	4	3	293	95.8	2.50×10^{-6}	1.06×10^{2}
4	4	1	269	105	1.91×10^{-6}	7.44×10^{1}
4	4	2	293	95.8	2.50×10^{-6}	1.06×10^{2}
4	4	3	296	89.6	2.60×10^{-6}	1.22×10^{2}
6	3	1	69.2	17.7	3.34×10^{-8}	3.67×10^{1}
6	3	2	255	64.8	1.68×10^{-6}	1.38×10^{2}
6	3	3	238	85.2	1.33×10^{-6}	7.51×10^{1}
6	3	4	273	110	2.01×10^{-6}	7.18×10^{1}
7	5.5	1	306	100	2.86×10^{-6}	1.11×10^{2}
7	5.5	2	281	89.6	2.22×10^{-6}	1.07×10^{2}
7	5.5	3	346	115	4.12×10^{-6}	1.22×10^{2}
7	5.5	4	252	80.7	1.60×10^{-6}	9.47×10^{1}
8	2	1	255	76.3	1.65×10^{-6}	1.06×10^{2}
8	2	2	297	113	2.59×10^{-6}	8.58×10^{1}
8	2	3	268	95.8	1.91×10^{-6}	8.48×10^{1}
8	2	4	287	90.5	2.35×10^{-6}	1.10×10^{2}
8	2	5	265	78.9	1.87×10^{-6}	1.12×10^{2}
9	5	1	285	89.6	2.31×10^{-6}	1.10×10^{2}
9	5	2	271	70.9	2.01×10^{-6}	1.40×10^{2}
9	5	3	292	82.5	2.50×10^{-6}	1.33×10^{2}

Table 9: Summary of bead-indentation measurements and computed Young's modulus.

In the above table C_{col} denotes the collagen concentration, R_{bead} the radius of the bead and δ the deformation of the gel (see Figure 8).